Advanced ABM Assignment 5

Ayush Sarkar

Due: April 15th, 2026

Part 1: Social Darwinism

Problem 1

Without any programming, what is the standard deviation of wealth at the start of this run (at Setup)? If the Social Darwinist assumption (only ability matters) were correct, would the standard deviation change in the course of a run? Explain your answer.

Answer 1:

Looking at the code, at setup the wealth distribution is defined as a random-float in [0,20], and if we assume the "randomness" to follow a uniform distribution, we can use the standard deviation formula for a uniform random variable such that:

$$\frac{(b-a)}{\sqrt{12}} = \frac{20}{\sqrt{12}} \approx 5.77$$

If we follow the Social Darwinism assumption that only the "fittest" individuals survive and prosper, we can see that only the agents with the best vision and lowest metabolism survive. We also can see that the total population curve follows an exponential decay curve, converging after about ≈ 160 ticks to a total of about ≈ 90 agents. Then running the simulation a few times, we can see the wealth distribution doesn't really converge to a definite known distribution, but is bounded below by some non-zero integer and bounded above by 500.

Now, if we update the metabolism, vision, and initial wealth of each agent to be fixed to 3, 3, and 20 respectively and enable the instant replenishment of sugar, the standard deviation of the initial wealth distribution is zero. Running the simulation, we can observe that the wealth distribution moderately resembles a normal distribution that travels upwards in the compact interval [0, 500]. Seeing this, we can assert that the initial location of an agent (under the given modifications to the settings) matters more to that individual agent's survival than any of their traits due. The normal distribution shape also seems to come from the fact that each agent acts purely out of self-interest and tries to move towards the most profitable patch in their vision while agents too far from patches with the bare minimum die off, causing any tails to disappear and the center of the distribution to travel upwards as the average wealth accumulates.

Problem 2

Do some agents die? Would this happen if Social Darwinism were true (that is, if agents are identical in both "ability" and initial sugar wealth, and birth location were irrelevant)? Explain your answer.

Answer 2:

Unfortunately, agents will die in either setup, as there is a finite amount of resources and space, and their survival is moreso dependent upon initial location (in the modified, fixed setting) and/or vision and metabolism (in the Social Darwinism setting). In either example, the limited amount of resources and the inequality of the number of agents relative to those resources causes at least some number of agents to die.

Problem 3

In the Histogram plot window, set $x_{\max} = 3000$. Submit a screenshot of the histogram of wealth at ticks = 3000. It starts with a vertical spike at the common initial wealth of 20. How does the histogram evolve from there? Does this evolving wealth histogram accord with the Social Darwinist prediction? Explain your answer.

Answer 3

Upon running the simulation multiple times under the fixed setup, we can observe that the wealth distribution actually seems to follow a right-skewed distribution, representing that a small subset of the population is wealthy, while the other percentiles lag behind. This is caused by the initial position of the agent and the fact that some agents die off, while the agents closer to the highest-producing patch are more easily able to access and profit off the sugar produced by the center patch. Thus, we can assert that the distribution becomes highly unequal and that the wealth inequality is driven by initial position, rather than any one agent being more "fit" than another.

Initial Wealth Distribution Screenshot

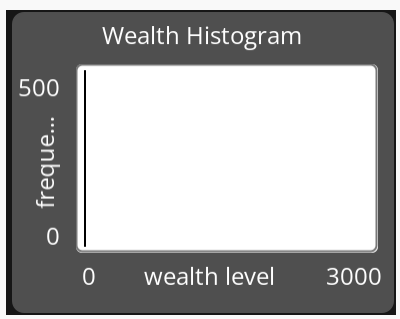

Figure 1: Initial Wealth Distribution in Modified Sugarscape Model

Final Wealth Distribution Screenshot

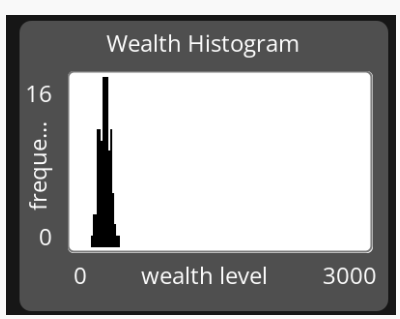

Figure 2: Final Wealth Distribution in Modified Sugarscape Model

Problem 4

Add a new plot window displaying the standard deviation of turtle wealth over time. Submit a screenshot of this new standard deviation plot from the starting time to the stopping time. True or False: To refute the Social Darwinian hypothesis, it would suffice for this curve to exceed zero. Explain your answer.

Answer 4

We can observe that the standard deviation of all the turtles' wealth grows from zero and increases as the number of ticks increases. This occurs because each turtle begins with an identical amount of wealth. Still, since their survival is determined by initial position and access to higher-producing patches, the distribution and its standard deviation reflect that reality through increasing amounts of variance in the total wealth distribution of all turtles.

Standard Deviation Plot

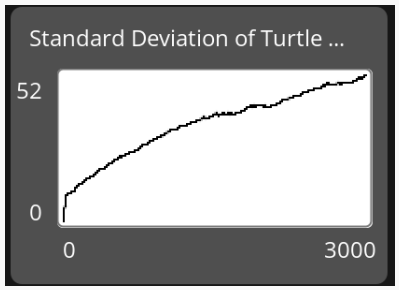

Figure 3: Evolution of Standard Deviation of Turtle Wealth

Problem 5

Characterize this curve: Is it rising, falling, or flat? Carefully explain why that is the behavior. It is not sufficient to say, for instance, "It is rising because the level grows with time." This simply restates the question. By what mechanism is this the case? Submit a screenshot of the spatial Sugarscape window at ticks = 3000.

Answer 5

We previously observed that the initial standard deviation is zero, as the initial wealth for all agents is fixed to zero. Then, when we run the simulation, we can see that the plot of standard deviation over time takes on a the shape of a rising curve. This comes as a consequence of the fact that an individual agent's survival is determined by their initial position and immediate access to higher-producing patches of sugar, and not their intrinsic traits like metabolism and vision. The death of agents far from the patches that at least keep up with their metabolism further perpetuates this inequality, as the rich get richer while the poor are too far from the necessary resources needed for their survival. Similarly, due the finite amount of resources, the agents with an initial position near patches that can at least, if not exceed, their metabolism begin crowding, leaving no room for the agents too far those same patches.

Sugarscape Spatial View

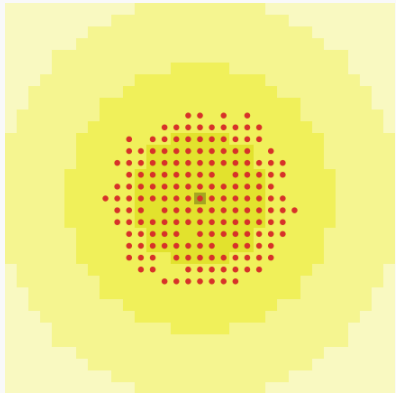

Figure 4: Final Sugarscape Spatial View w/ Fixed Settings

Problem 6

In light of this exercise, please discuss the following assertion: Completely unrealistic "toy" models, using no real-world data, can sometimes be used to call social theories into question.

Answer 6

True: The Sugarscape model and the results seen above challenge the theory of Social Darwinism, as when we fix the initial traits and wealth of all agents to be equal, we can still observe the proliferation of wealth disparity and can isolate the cause of that disparity to the initial position of the agent rather than any lack of ability or individual difference. When we add heterogeneity to the initial traits and wealth of the agents, we don't see the same wealth inequality appear, and so, we can directly sever the claim that the "survival of the fittest" is caused by "fitness", and rather, it is caused by unequal access to opportunity. Beyond the Sugarscape model and Social Darwinism, we can help demonstrate that certain claims of causality behind social, economic, and health-related outcomes are incorrect by having simulations ran in vacuum with no confounders.

Part 2: Sugarscape Modifications

Problem 7

Open the Diagonal Waves model (2020JScape2peakDiagonalWaves.nlogo). Select continuous, not terraces. Select instant growback. Does this change the diagonal waves result? If so, explain why.

Answer 7

Now running the diagonal waves NetLogo code with continuous gradients of sugar patches and instant growth, we can observe that the diagonal waves that originally appeared on the Terrace topography with constant sugar growth produces clear diagonal wavefronts that propogate towards the second terrace of available resources. Unlike the simulation under the original settings, the continuous topography and instant growth of sugar patches produces a different pattern, one where the primary terrace becomes fully saturated with agents. Due to this saturation, eventually there is too little space and a few of the agents from the original terrace stay outwards and stumble upon the second terrace, which we can see in the pictures below.

Diagonal Waves

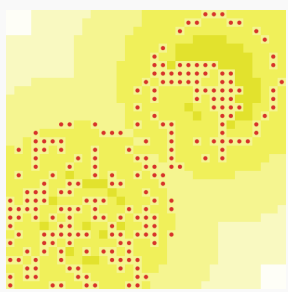

Figure 5: Diagonal Waves on Terraces w/ Constant Growth

Stray Agent

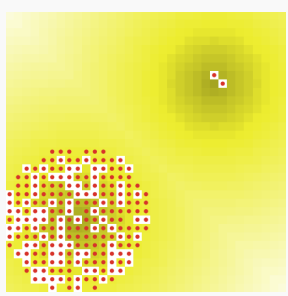

Figure 6: Stray Agent on Continuous Topography w/ Instant Growth

Problem 8

1. **(a)** Expand the world to 32×32. Increase the population to 1000. Add a third peak centered at $(-15, 15)$. Show a screenshot of this landscape.

2. **(b)** Set up the turtles so that initially they occupy random positions with $pxcor < -15$. Their *pycor* values remain random as before. Show a screenshot of this configuration.

3. **(c)** True or False: With instant growback, they make it to all three peaks. Explain the result.

Answer 8

False: The agents do not make it to all three terraces, as enabling instant growth of the sugar patches remove the agents' need to explore and find resources to survive. This is related to the explore-exploit tradeoff, as when resources are finite and temporally limited, the agents will need to expand outwards to survive. Likewise, when there isn't an issue of resources or growback rates, the agents' needs are fulfilled by the first two terraces, and so, they don't need to explore outwards and therefore don't expand to the third terrace.

(Part (a): Landscape Screenshot

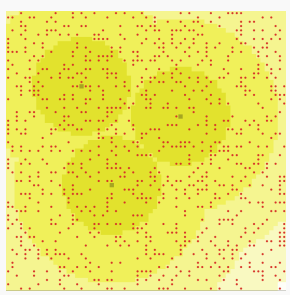

Figure 7: Updated Spatial Landscape (32×32 w/ third terrace)

Updated Initial Position (Part (b)

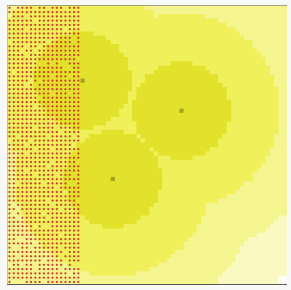

Figure 8: Updated Agent Position ($x < -15$)

(Part (c): Simulation of Agent Movement w/ Three Terraces and Updated Initial Position (Constant Growth)

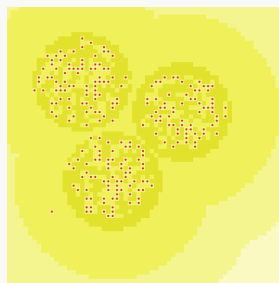

Figure 9: Asymptotic Movement of Agents w/ Updated Setup (Constant Growth)

Part (c): Simulation of Agent Movement w/ Three Terraces and Updated Initial Position (Instant Growth)

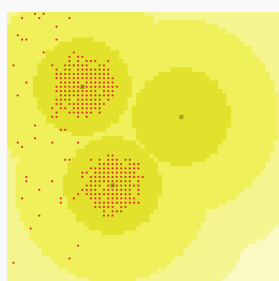

Figure 10: Asymptotic Movement of Agents w/ Updated Setup (Instant Growth)